

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \dagger \quad \mathcal{K}_{0^+}^{\#2} \\
\begin{array}{|c|c|}
\hline
\frac{1}{2} (-2 \Upsilon_1 k^2 + \overset{(2)}{\kappa_1}) & \frac{1}{2} (2 \Upsilon_1 k^2 - \overset{(2)}{\kappa_1}) \\
\hline
\frac{1}{2} (2 \Upsilon_1 k^2 - \overset{(2)}{\kappa_1}) & \frac{1}{2} (-2 \Upsilon_1 k^2 + \overset{(2)}{\kappa_1}) \\
\hline
\end{array} \\
\mathcal{K}_{1^+}^{\#1} \quad \mathcal{K}_{1^+}^{\#2} \\
\begin{array}{|c|c|}
\hline
\mathcal{K}_{1^+}^{\#1} \dagger^\alpha & \frac{1}{6} (-10 \Upsilon_2 k^2 + 5 \overset{(2)}{\kappa_1}) & \frac{1}{6} (2 \sqrt{5} \Upsilon_2 k^2 - \sqrt{5} \overset{(2)}{\kappa_1}) \\
\hline
\mathcal{K}_{1^+}^{\#2} \dagger^\alpha & \frac{1}{6} (2 \sqrt{5} \Upsilon_2 k^2 - \sqrt{5} \overset{(2)}{\kappa_1}) & \frac{1}{6} (-2 \Upsilon_2 k^2 + \overset{(2)}{\kappa_1}) \\
\hline
\end{array} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \mathcal{K}_{2^+}^{\#1} \alpha\beta & -2 \Upsilon_3 k^2 + \overset{(2)}{\kappa_1} \\
\mathcal{K}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \mathcal{K}_{3^+}^{\#1} \alpha\beta\chi & -2 \Upsilon_4 k^2 + \overset{(2)}{\kappa_1}
\end{array}$$

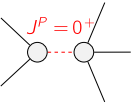
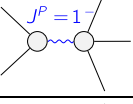
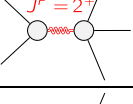
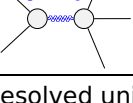
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\begin{array}{|c|c|}
\hline
\mathcal{J}_{0^+}^{\#1} \dagger & \frac{1}{2 \text{Det}(0^+)} & -\frac{1}{2 \text{Det}(0^+)} \\
\hline
\mathcal{J}_{0^+}^{\#2} \dagger & -\frac{1}{2 \text{Det}(0^+)} & \frac{1}{2 \text{Det}(0^+)} \\
\hline
\end{array} \\
\mathcal{J}_{1^+}^{\#1} \quad \mathcal{J}_{1^+}^{\#2} \\
\begin{array}{|c|c|}
\hline
\mathcal{J}_{1^+}^{\#1} \dagger^\alpha & \frac{5}{6 \text{Det}(1^+)} & -\frac{\sqrt{5}}{6 \text{Det}(1^+)} \\
\hline
\mathcal{J}_{1^+}^{\#2} \dagger^\alpha & -\frac{\sqrt{5}}{6 \text{Det}(1^+)} & \frac{1}{6 \text{Det}(1^+)} \\
\hline
\end{array} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \mathcal{J}_{2^+}^{\#1} \alpha\beta & \frac{1}{\text{Det}(2^+)} \\
\mathcal{J}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \mathcal{J}_{3^+}^{\#1} \alpha\beta\chi & \frac{1}{\text{Det}(3^+)}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
\Upsilon_1 &= 7 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \& \Upsilon_2 = 6 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \& \Upsilon_3 = 4 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \& \Upsilon_4 = \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \& \\
\text{Det}(0^+) &= -2 k^2 (7 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) + \overset{(2)}{\kappa_1} \& \text{Det}(1^+) = -2 k^2 (6 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) + \overset{(2)}{\kappa_1} \& \\
\text{Det}(2^+) &= -2 k^2 (4 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) + \overset{(2)}{\kappa_1} \& \text{Det}(3^+) = -2 k^2 (\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) + \overset{(2)}{\kappa_1}
\end{aligned}$$

Lagrangian

$$\begin{aligned}
& \overset{(2)}{\kappa_1} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - \frac{1}{2} \overset{(2)}{\kappa_1} \mathcal{K}^\alpha{}_\beta \mathcal{K}^\beta{}_\chi + \overset{(4)}{\kappa_1} \partial_\beta \mathcal{K}_\chi{}^\delta \partial^\chi \mathcal{K}_\alpha{}^\beta + \overset{(4)}{\kappa_2} \partial_\chi \mathcal{K}_\beta{}^\delta \partial^\chi \mathcal{K}_\alpha{}^\beta - \\
& 18 \overset{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + 12 \overset{(4)}{\kappa_1} \partial^\chi \mathcal{K}_\alpha{}^\beta \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 2 \overset{(4)}{\kappa_1} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\kappa_2} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#2} + 3 \mathcal{J}_{0^+}^{\#1} = 0$	1	$\partial_\beta \mathcal{J}^\alpha{}_\beta = 0$	
$\mathcal{J}_{1^+}^{\#2\alpha} + \mathcal{J}_{1^+}^{\#1\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi = \partial_\chi \partial^\chi \mathcal{J}^\alpha{}_\beta$	
Total # constraint(s):	4		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$\frac{\overset{(2)}{\kappa_1}}{2 (7 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2})}$	$-\frac{1}{2 (7 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2})}$
	3	$\frac{\overset{(2)}{\kappa_1}}{2 (6 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2})}$	$\frac{1}{12 \overset{(4)}{\kappa_1} + 2 \overset{(4)}{\kappa_2}}$
	5	$\frac{\overset{(2)}{\kappa_1}}{8 \overset{(4)}{\kappa_1} + 2 \overset{(4)}{\kappa_2}}$	$-\frac{1}{2 (4 \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2})}$
	7	$\frac{\overset{(2)}{\kappa_1}}{2 (\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2})}$	$\frac{1}{2 (\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2})}$
Resolved unitarity condition(s):		(Demonstrably impossible)	